Evaluating Protection Data Model Quality in Gridscale X Advanced Protection Assessment

by

Grid Integrity Solutions, LLC

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Evaluating Protection Data Model Quality

This document describes an automated workflow for evaluating protection data model quality in Gridscale X APA, using APA macros and a Power BI reporting tool.

For first-time setup, proceed directly to the Quick Start section below (Section 1).

1 Quick Start – setup and first run

This section provides a step-by-step guide to set up and run the protection data quality checking workflow. Follow these steps in the order provided for a successful setup.

✅ Step 1 – Download the APA Macros and Power BI Report Files

1. Download the Git repository at https://github.com/Grid-Integrity-Solutions/protection-qc
2. Extract the macros and Power BI Report Files

✅ Step 2 – Install the APA Macro Files

1. Copy the following macro files into your APA user folder:
    a. **perform_protection_data_check_and_report.mac**
    b. **perform_protection_data_check_and_report_utils1.mac**
2. Typical APA user folder locations include:
    a. **C:\cape\dat**
    b. **C:\Users\<your username>\APA16User**
3. If unsure, consult your APA or IT administrator.

✅ Step 3 – Update the APA Configuration File

1. Open the file **cape.cfg** (located in your APA user folder) using a text editor.
2. Locate the section labeled **One-Line Diagram Module**
3. Add the following line just above the **`return`** statement:
    a. **`input perform_protection_data_check_and_report.mac`**

```
%%%
```

✅ Step 6 – Verify CSV Output Files

After the macro completes:

1. Navigate to your APA user folder.
2. Confirm that the following folders have been created:
    a. **PPDCR_Output_Files\Sub_Output**
    b. **PPDCR_Output_Files\Ele_Output**
3. Verify that CSV files are present in both folders.

✅ Step 7 – Load Data into Power BI

1. Open the Power BI report **Protection_Data_Quality_Check.pbix**
2. Click: **Transform Data**
3. Select: **Manage Parameters → Edit Parameters**
4. Update the parameter: **APA_User_Dir**
    a. Example: **C:\Users\<your username>\APA16User**
    b. This needs to be done only the first time the report is opened. Be sure to save the report before exiting Power BI so that the **APA_User_Dir** folder path is remembered by the report.
5. Click **OK**, then click:
    a. Close & Apply
6. Wait for the report to refresh. This could take some time depending on the amount of data that needs to be processed.

✅ Step 8 – Verify Successful Setup

The setup is complete if:

- ✅ Data appears in the **Line/XFMR Terminal Summary** page
- ✅ Tables and charts are populated
- ✅ APA command values (e.g., VIEW_LZOP_DATA xxxx) are visible
- ✅ Asset count is greater than zero
- ✅ Data will appear in the **Element-Level Details** page as well (assuming the LZOPs are populated with relays containing elements with contact logic codes)

⚠️ Troubleshooting

If you encounter issues, try the checks below.

❌ No Data Appears in Power BI

- Verify that the **APA_User_Dir** parameter is correct and matches your installation of APA
- Ensure CSV files exist in:
  - **PPDCR_Output_Files\Sub_Output**
  - **PPDCR_Output_Files\Ele_Output**
- Confirm you clicked **Close & Apply** after updating the **APA_User_Dir** parameter

❌ CSV Files Not Found

- Confirm the macro ran successfully in APA
- Verify the macro files are:
  - correctly copied
  - correctly referenced in **cape.cfg**

❌ Macro Not Visible in ShortCuts Menu

- Verify:
  - **cape.cfg** was updated correctly
  - **ol_shortcuts.txt** contains the correct entries
- Restart APA after making changes

❌ APA Command Column is Blank

- The selected asset may not have a valid LZOP assigned
- This is expected for certain data conditions (missing LZOP for an asset)

❌ Errors During Power BI Refresh

- Check that the folder path exists and is accessible
- Ensure CSV files are not open in another application (e.g., Excel)
- Confirm file naming and folder structure have not been altered

✅ Still Need Help?

Contact: Grid Integrity Solutions; info@gridintegritysolutions.com

✅ Tip

For best results:

- Re-run the APA macro whenever data changes
- Refresh Power BI after each macro run
- Use filters and visuals to prioritize high-severity issues

✅ Result

After completing this workflow, you will have:

- A consolidated view of protection data quality issues
- Clear identification of actionable fixes
- Direct navigation from Power BI to APA data forms via command-line integration

If you are able to successfully complete the Quick Start steps above, you can begin using the report immediately.

For additional details, screenshots, and explanations of each step, refer to the sections below.

2 Introduction

The protection data model in Siemens Gridscale X Advanced Protection Assessment (APA) software comprises several components:

1. Local Zone of Protection (LZOP) – container to hold relays that protect a certain primary network object, and trip open an associated set of circuit breakers
2. Protective relays
3. Current transformers (CTs) and Voltage (or Potential) transformers (VTs) connected to the relays
4. Contact logic codes that combine the outputs of the various protection elements within the relays and enable the LZOP to trip open the associated circuit breakers

Entering the protection data into the APA database and preparing the model for performing protection coordination studies is a significant effort that the modeling team at a company must undertake to ensure that they get the most benefit from the advanced simulation capabilities in APA.

Errors in the protection data model can reduce the quality of the results obtained from running detailed protection coordination studies. Some of these errors are:

1. Duplicate contact logic codes on protection elements in the same LZOP: This error will cause APA to exclude such elements from the simulation.
2. Missing CTs or CT location mismatch: Absence of a CT connection will prevent an element from operating for a fault condition. Connecting a protection element to a CT from a branch other than the protected branch could cause it to misoperate for a fault.
3. Missing VTs or VT location mismatch: Absence of a VT connection will prevent an element from operating for a fault condition. Connecting a protection element to a VT from a bus or branch that does not match the protected branch could cause it to misoperate for a fault.
4. Missing LZOP logic expressions: Absence of a logic expression at the LZOP level will prevent circuit breaker tripping even if the LZOP includes relays with no modeling errors.

These and other errors such as inverse-time overcurrent math model errors, unused logic codes and out of range settings on elements can all adversely affect the quality of the simulations performed by creating false positives (not a real protection settings issue) or failing to capture real protection vulnerabilities that need mitigation.

Gridscale X APA provides its users with tools to check the quality of the protection data and identify the errors that can reduce the quality and fidelity of the protection coordination studies that are performed. These tools produce output in the form of text messages and

information in log files, which the user must parse to understand the modeling issue, and take the necessary steps to resolve it.

This document describes a more automated data quality checking process (compared to APA's current method) that provides consolidation and tabulation of data checking, with clear visualization of data quality issues. This automated approach provides the user with an unambiguous listing of actionable items, which when resolved will improve the data model quality, making it ready for protection studies.

We start with a brief description of the data checking tools that are currently available in APA.

3 Protection data checking in APA (current method)

The usual process for running a protection data quality check in APA is to set the **Reporting Level** in the System Simulator or Relay Checking modules to **Data_Check_Only** or **Data_Check_Errors_Only** and run a fault clearing simulation. See Figure 1.

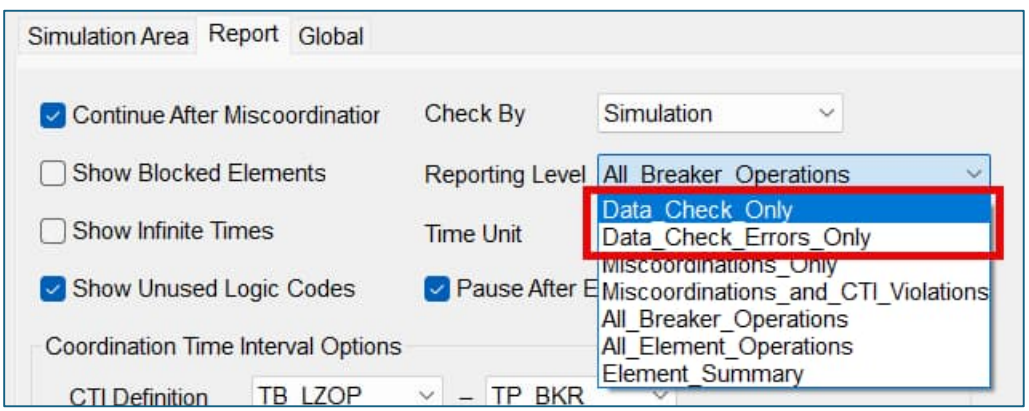

Figure 1: Set Reporting Level for data checking.

The result of the simulation will produce text output in the text area of APA. See Figure 2.

```
Simulation Area:
   130 buses;   141 branches
LINE LZOP                  90 lzops
MISC LZOP                   4 lzops
 94 distinct lzops
There are missing taps detected! Please check |ApaUserDir|\cape_15_log.txt for more details.
In Substation "SUB_8" LZOP 4396 LZOP_4396
LZOP #4396 has duplicated contact logic code "51G_1". Tripping suppressed for this LZOP

Errors/Warnings while Reading Data and Building Simulation
----------------------------------------------------------
There are missing taps detected! Please check |ApaUserDir|\cape_15_log.txt for more details.
In Substation "SUB_8" LZOP 4396 LZOP_4396
LZOP #4396 has duplicated contact logic code "51G_1". Tripping suppressed for this LZOP



                              TOC Curve Validation
                            ----------------------------

No TOC curve mismatches found.

Data verification completed
```

Figure 2: Data checking output in the text area.

The text area output, in combination with data in the log file produced by APA, must be parsed and understood by the user. Then, the user must access the appropriate data form in APA to locate the error and fix it. Figure 3 summarizes this process.

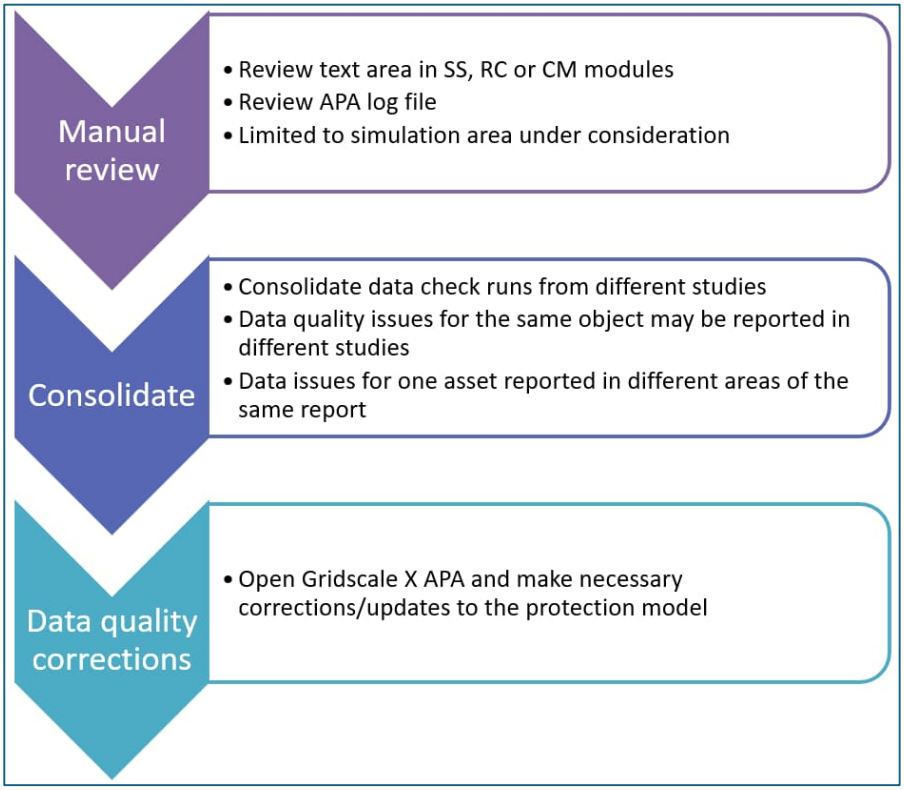

Figure 3: Summary of current data checking process in APA.

4 Enhanced protection data checking

The enhanced protection data checking method provides the following advantages over the current methods available in APA:

1. A more automated process using a special APA macro
2. Consolidation of issues found in multiple data-checking runs
3. Data quality issues for the same asset (reported in different areas of the text output) now combined into one report for that asset
4. Clear list of action items that the user needs to resolve, via a Power BI report

The workflow for the enhanced method is shown in Figure 4.

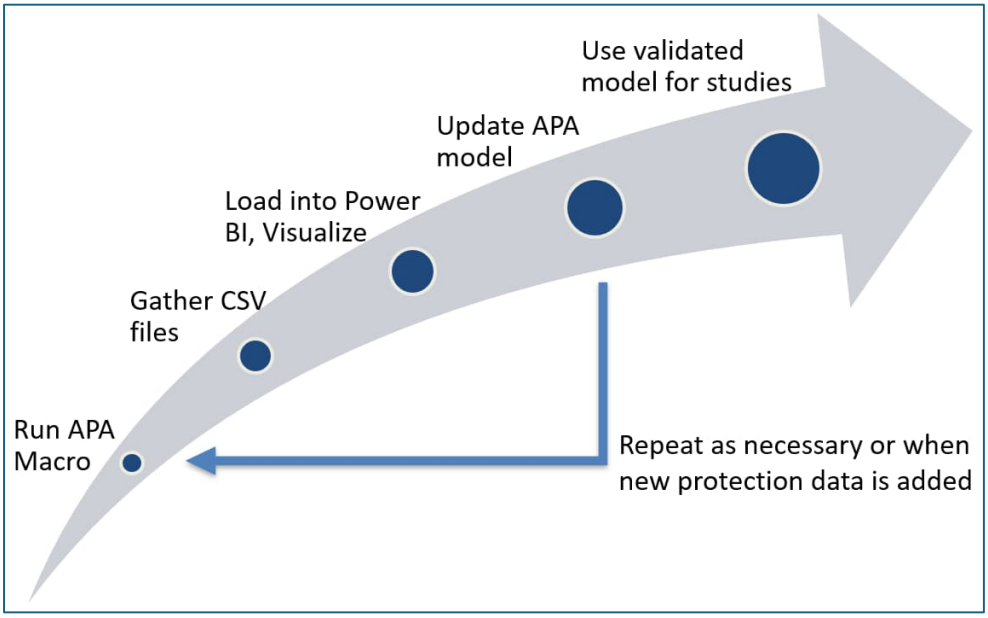

Figure 4: Workflow for enhanced protection data quality checking.

The various steps of the workflow along with detailed setup and usage instructions are elucidated in the sections that follow.

5 Detailed setup and usage

5.1 APA macro installation

The new APA macro for protection data quality checking consists of two (plain text) files:

- **perform_protection_data_check_and_report.mac**
- **perform_protection_data_check_and_report_utils1.mac**

To install the macros:

1. Copy the macro files (provided by Grid Integrity Solutions) into a suitable folder – typically, your APA user folder. Some possibilities for the user folder location are (depending on APA version and your installation):
   - **C:\cape\dat**
   - **C:\Users\<myusername>\APA16User**

   Please check with your APA administrator or IT administrator for details.

5.2 APA configuration file update

To enable the new macro to be called from the ShortCuts menu, you will need to update the user configuration file called **cape.cfg**.

1. Locate **cape.cfg** in the same APA user folder discussed above. Open it in a text editor like Notepad, Notepad++ or other program.
2. Locate the section in this file labeled **One-Line Diagram Module**
3. Enter the line **input perform_protection_data_check_and_report.mac** just above the **return** statement. See the highlighted text below.

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%One-Line Diagram Module %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
[OL]
%Auto Overwrite Area Begins
%Last modified by "xxxxx" on May 31 2026 at 20:41:30
%Auto Overwrite Area Ends

input perform_protection_data_check_and_report.mac
return
```

4. Save the **cape.cfg** file. Including this statement in the **cape.cfg** file ensures that the macro is available and ready to run when you start APA for your work.

Please note that the One-Line Diagram Module is used for running the data quality check because protection data entry is typically carried out from that module. You can certainly run the macro from another module in APA if you wish to do so. Simply move the **input** statement to the appropriate section of the **cape.cfg** file.

5.3 APA ShortCuts menu update

Perform the following actions to add a menu item to the **ShortCuts** menu in the One-Line Diagram Module:

1. If your APA user folder already contains a file with the name **ol_shortcuts.txt**, open that file in the text editor.
2. If the file does not exist, create a new one.
3. Then, (in the existing file or new one) add the following lines:

```
Protection Data Checking
  perform_protection_data_check_and_report
```

Note that the second line in shown above is indented by two spaces from the first to provide a suitable menu structure.

A sample **ol_shortcuts.txt** file has been provided along with the other files for your convenience. If running the macro in a module other than One-Line, use the appropriate

two-letter abbreviation for the shortcuts file; for example, **ss_shortcuts.txt** if using the System Simulator module. The content remains the same.

5.4 Running the macro in APA

To run the new data checking macro in APA:

1. Start APA, attach your database and build the short-circuit network model.
2. Change to the One-Line Diagram module.
3. Click the **ShortCuts** menu.
4. If the actions described in Sections 5.2 and 5.3 have been performed correctly, you will see the menu selection shown in Figure 5.
5. Select **perform_protection_data_check_and_report**.

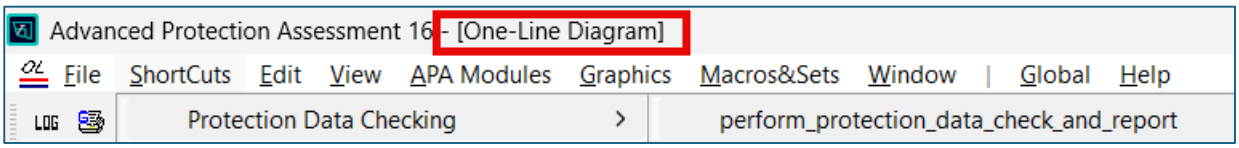

Figure 5: Running the protection data quality checking macro.

You will be presented with the pop-up shown in Figure 6. Enter/select data in this form:

1. Run the study on:
    a. One line or transformer (or)
    b. All lines and transformers in a substation (or)
    c. All substations in the database
2. Specific line or transformer (depends on selection in Q1)
3. Specific substations (depends on selection in Q1)
4. Threshold voltage in kV below which equipment terminals will be ignored by the macro (excluded from data checking)
5. The bus area(s) you want to restrict the data checking to:
    a. Select “0-None” if you don’t want to restrict the data checking to a particular area
    b. You can select up to 5 of the bus areas presented on the form.
    c. Some APA databases may not have any bus areas defined in them. In such cases, the area selection options will not appear.

Then, click **Ok** to run the macro.

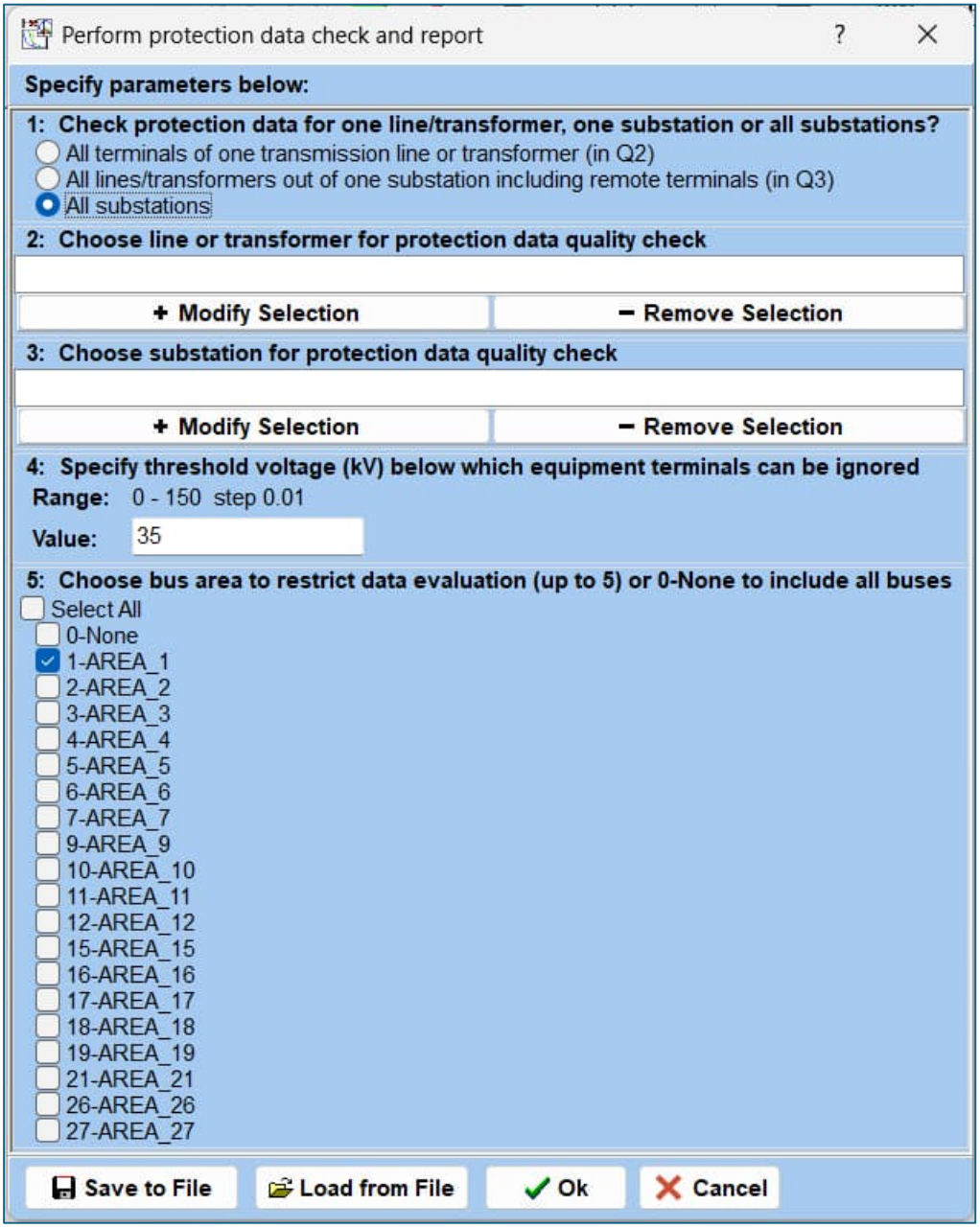

Figure 6: Macro selections.

5.5 Data checks performed by the APA macro

An APA macro does not have access to all the internal data structures within APA. Therefore, this first version of the data checking macro performs the following tests, which are easier to implement using existing calls in CUPL. Additional tests will be implemented and issued via future updates to the macro.

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- LZOP-level data quality checks
  - Missing LZOP on a line or transformer terminal
  - Missing LZOP logic expression (if LZOP is present)
- Relay-level data quality checks
  - Missing relays in an LZOP
  - Absence of elements with contact logic codes
- Element-level data quality checks
  - Duplicated element contact logic codes within an LZOP
  - Missing operating CT connection for IOC, TOC and DIST elements
  - CT location mismatch
  - Missing operating VT connection for DIST elements
  - VT location mismatch

In this version, the APA macro will select IOC, TOC, DIST, TIMER, CDIFF and AUX elements for evaluation. These element types must have either the trip CB flag set to "Y" or equipped with a non-blank contact logic code.

5.6 Screen output produced by the APA macro

As the macro performs its data quality tests, it will produce output on the screen in APA. Sample output is shown in Figure 7, Figure 8 and Figure 9.

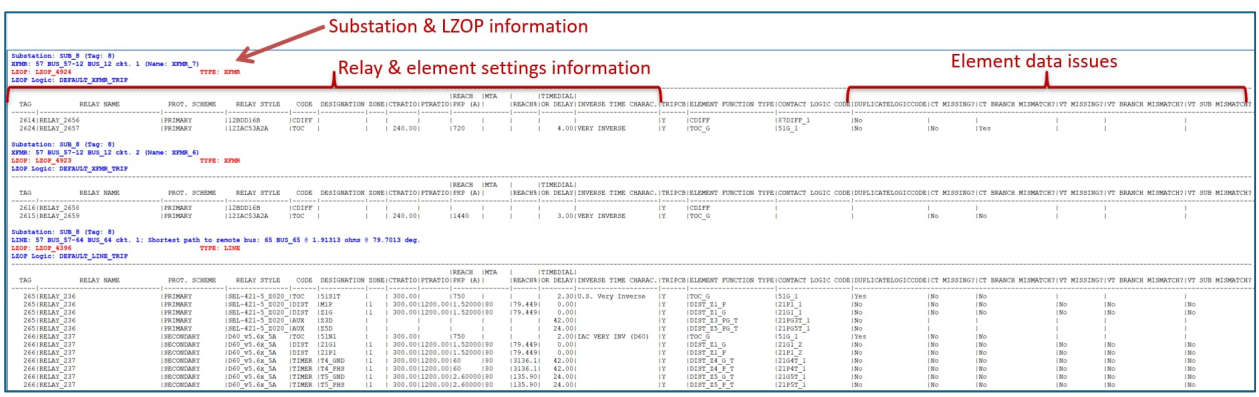

Figure 7: Screen text showing substation, LZOP, relay, settings and element data issues.

The screen output (Figure 8) shows details about the protective relay elements, including relay name, relay style, element CT and PT ratios, reach or pickup setting, and other relevant data. This gives the user a quick overview of the element-level settings.

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Relay & element settings information

```
Substation: SUB_8 (Tag: 8)
XFMR: 57 BUS_57-12 BUS_12 ckt. 1 (Name: XFMR_7)
LZOP: LZOP_4924                                   TYPE: XFMR
LZOP Logic: DEFAULT_XFMR_TRIP
-----------------------------------------------------------------------------------------------------------------------------------------------------
                                                                                                  |REACH  |MTA   |      |TIMEDIAL|
  TAG            RELAY NAME            PROT. SCHEME     RELAY STYLE    CODE  DESIGNATION ZONE|CTRATIO|PTRATIO|PKP (A)|      |REACH%|OR DELAY|INVERSE TIME CHARAC.|
------|-----------------------------|---------------|---------------|------|-----------|----|-------|-------|-------|------|------|--------|--------------------|
  2614|RELAY_2656                   |PRIMARY        |12BDD16B       |CDIFF |           |    |       |       |       |      |      |        |                    |
  2624|RELAY_2657                   |PRIMARY        |12IAC53A2A     |TOC   |           |    | 240.00|       |720    |      |      |    4.00|VERY INVERSE        |

Substation: SUB_8 (Tag: 8)
XFMR: 57 BUS_57-12 BUS_12 ckt. 2 (Name: XFMR_6)
LZOP: LZOP_4923                                   TYPE: XFMR
LZOP Logic: DEFAULT_XFMR_TRIP
-----------------------------------------------------------------------------------------------------------------------------------------------------
                                                                                                  |REACH  |MTA   |      |TIMEDIAL|
  TAG            RELAY NAME            PROT. SCHEME     RELAY STYLE    CODE  DESIGNATION ZONE|CTRATIO|PTRATIO|PKP (A)|      |REACH%|OR DELAY|INVERSE TIME CHARAC.|
------|-----------------------------|---------------|---------------|------|-----------|----|-------|-------|-------|------|------|--------|--------------------|
  2616|RELAY_2658                   |PRIMARY        |12BDD16B       |CDIFF |           |    |       |       |       |      |      |        |                    |
  2615|RELAY_2659                   |PRIMARY        |12IAC53A2A     |TOC   |           |    | 240.00|       |1440   |      |      |    3.00|VERY INVERSE        |

Substation: SUB_8 (Tag: 8)
LINE: 57 BUS_57-64 BUS_64 ckt. 1; Shortest path to remote bus: 65 BUS_65 @ 1.91313 ohms @ 79.7013 deg.
LZOP: LZOP_4396                                   TYPE: LINE
LZOP Logic: DEFAULT_LINE_TRIP
-----------------------------------------------------------------------------------------------------------------------------------------------------
                                                                                                  |REACH  |MTA   |      |TIMEDIAL|
  TAG            RELAY NAME            PROT. SCHEME     RELAY STYLE    CODE  DESIGNATION ZONE|CTRATIO|PTRATIO|PKP (A)|      |REACH%|OR DELAY|INVERSE TIME CHARAC.|
------|-----------------------------|---------------|---------------|------|-----------|----|-------|-------|-------|------|------|--------|--------------------|
   265|RELAY_236                    |PRIMARY        |SEL-421-5_Z020_|TOC   |51S1T      |    | 300.00|       |750    |      |      |    2.30|U.S. Very Inverse   |
   265|RELAY_236                    |PRIMARY        |SEL-421-5_Z020_|DIST  |M1P        |1   | 300.00|1200.00|1.52000|80    |79.449|    0.00|                    |
   265|RELAY_236                    |PRIMARY        |SEL-421-5_Z020_|DIST  |Z1G        |1   | 300.00|1200.00|1.52000|80    |79.449|    0.00|                    |
   265|RELAY_236                    |PRIMARY        |SEL-421-5_Z020_|AUX   |Z3D        |    |       |       |       |      |      |   42.00|                    |
   265|RELAY_236                    |PRIMARY        |SEL-421-5_Z020_|AUX   |Z5D        |    |       |       |       |      |      |   24.00|                    |
   266|RELAY_237                    |SECONDARY      |D60_v5.6x_5A   |TOC   |51N1       |    | 300.00|       |750    |      |      |    2.00|IAC VERY INV (D60)  |
   266|RELAY_237                    |SECONDARY      |D60_v5.6x_5A   |DIST  |21G1       |1   | 300.00|1200.00|1.52000|80    |79.449|    0.00|                    |
   266|RELAY_237                    |SECONDARY      |D60_v5.6x_5A   |DIST  |21P1       |1   | 300.00|1200.00|1.52000|80    |79.449|    0.00|                    |
   266|RELAY_237                    |SECONDARY      |D60_v5.6x_5A   |TIMER |T4_GND     |1   | 300.00|1200.00|60     |80    |3136.1|   42.00|                    |
   266|RELAY_237                    |SECONDARY      |D60_v5.6x_5A   |TIMER |T4_PHS     |1   | 300.00|1200.00|60     |80    |3136.1|   42.00|                    |
   266|RELAY_237                    |SECONDARY      |D60_v5.6x_5A   |TIMER |T5_GND     |1   | 300.00|1200.00|2.60000|80    |135.90|   24.00|                    |
   266|RELAY_237                    |SECONDARY      |D60_v5.6x_5A   |TIMER |T5_PHS     |1   | 300.00|1200.00|2.60000|80    |135.90|   24.00|                    |
```

Figure 8: Relay and element settings details.

CT branch mismatch

```
-----------------------------------------------------------------------------------------------------------------------------------------------------
TRIPCB|ELEMENT FUNCTION TYPE|CONTACT LOGIC CODE|DUPLICATELOGICCODE|CT MISSING?|CT BRANCH MISMATCH?|VT MISSING?|VT BRANCH MISMATCH?|VT SUB MISMATCH?
------|---------------------|------------------|------------------|-----------|-------------------|-----------|-------------------|-----------------
Y     |CDIFF                |87DIFF_1          |No                |           |                   |           |                   |
Y     |TOC_G                |51G_1             |No                |No         |Yes                |           |                   |
```

```
-----------------------------------------------------------------------------------------------------------------------------------------------------
TRIPCB|ELEMENT FUNCTION TYPE|CONTACT LOGIC CODE|DUPLICATELOGICCODE|CT MISSING?|CT BRANCH MISMATCH?|VT MISSING?|VT BRANCH MISMATCH?|VT SUB MISMATCH?
------|---------------------|------------------|------------------|-----------|-------------------|-----------|-------------------|-----------------
Y     |CDIFF                |                  |                  |           |                   |           |                   |
Y     |TOC_G                |                  |                  |No         |No                 |           |                   |
```

Duplicate contact logic codes

```
-----------------------------------------------------------------------------------------------------------------------------------------------------
TRIPCB|ELEMENT FUNCTION TYPE|CONTACT LOGIC CODE|DUPLICATELOGICCODE|CT MISSING?|CT BRANCH MISMATCH?|VT MISSING?|VT BRANCH MISMATCH?|VT SUB MISMATCH?
------|---------------------|------------------|------------------|-----------|-------------------|-----------|-------------------|-----------------
Y     |TOC_G                |51G_1             |Yes               |No         |No                 |           |                   |
Y     |DIST_Z1_P            |21P1_1            |No                |No         |No                 |No         |No                 |No
Y     |DIST_Z1_G            |21G1_1            |No                |No         |No                 |No         |No                 |No
Y     |DIST_Z3_PG_T         |21PG3T_1          |No                |           |                   |           |                   |
Y     |DIST_Z5_PG_T         |21PG5T_1          |No                |           |                   |           |                   |
Y     |TOC_G                |51G_1             |Yes               |No         |No                 |           |                   |
Y     |DIST_Z1_G            |21G1_2            |No                |No         |No                 |No         |No                 |No
Y     |DIST_Z1_P            |21P1_2            |No                |No         |No                 |No         |No                 |No
Y     |DIST_Z4_G_T          |21G4T_1           |No                |No         |No                 |No         |No                 |No
Y     |DIST_Z4_P_T          |21P4T_1           |No                |No         |No                 |No         |No                 |No
Y     |DIST_Z5_G_T          |21G5T_1           |No                |No         |No                 |No         |No                 |No
Y     |DIST_Z5_P_T          |21P5T_1           |No                |No         |No                 |No         |No                 |No
```

Figure 9: Element data quality issues.

Figure 9 highlights some of the data quality issues detected by the macro for some of the protective relay elements.

5.7 CSV files produced by the macro

In addition to the screen output described in Section 5.6, the APA macro also produces comma-separated variable (CSV) files with relevant information for subsequent ingestion into the Power BI report. These files are produced in the APA user folder. See Figure 10.

In this example, the user folder is **APA16User**. Under this folder, the macro will create a top-level folder called **PPDCR_Output_Files**. Within this folder, two additional folder are created:

- **Sub_Output**: To store a CSV file with asset-level (line or transformer terminal) data
- **Ele_Output**: To store a CSV file with element-level data

Note that each CSV file in Figure 10 is time-stamped with the date and time when the macro was run. Furthermore, each line of data within both files is time-stamped with the same date and time to ensure that data quality issues can be tracked over time, when visualized in Power BI.

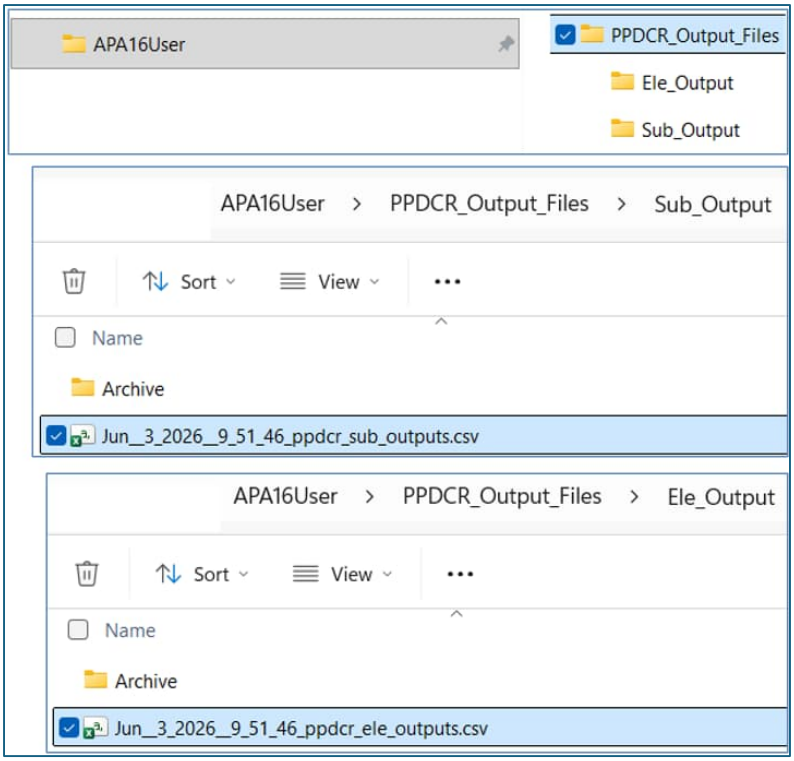


Figure 10: CSV files produced by the APA macro.

6 Power BI report walkthrough

The two CSV files produced by the APA macro must be ingested by the Power BI report for proper visualization of the data quality issues.

The report file provided by Grid Integrity is called **Protection_Data_Quality_Check.pbix**.

1. Copy the file to a suitable location on your computer. The APA user folder where the macro files are present is a convenient location for the Power BI report as well.

6.1 Ingesting the CSV files produced by the APA macro

1. Open the Power BI report file **Protection_Data_Quality_Check.pbix**.
2. You will need to have the desktop version of Power BI installed on your computer. The license for the desktop version is provided by Microsoft free of charge.
3. However, please check with your IT administrator to confirm that your company policy allows you to install and run Power BI desktop.
4. The report will open the **Quick Setup Guide** page, which summarizes the instructions for visualizing the data. We have already run the APA macro and created the CSV files. The next step is to load those files into the Power BI report.
5. Click the **Transform data** button. A new window will open. See Figure 11.

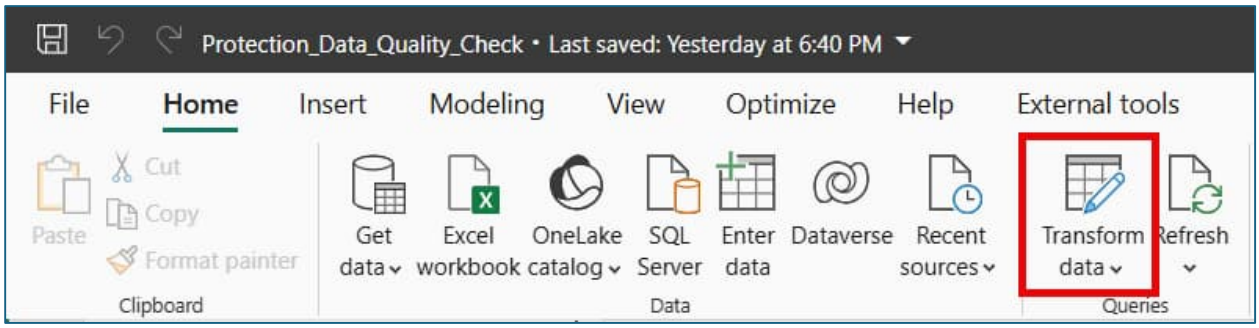


Figure 11: Click Transform data.

6. Click **Manage Parameters | Edit Parameters**. See Figure 12.

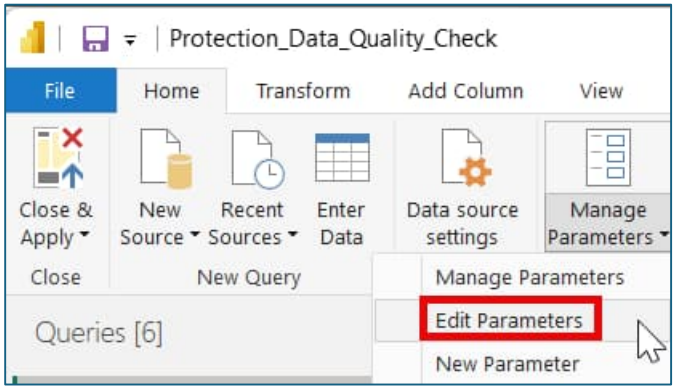


Figure 12: Click Edit Parameters.

7. In the resulting Edit Parameters popup (Figure 13), enter the full path to your APA user folder. Then, click **OK**.

Figure 13: Specify full path to the APA user folder.

8. Finally, click the **Close & Apply** button – Power BI will now read the CSV files and populate the various visuals in the report. See Figure 14.

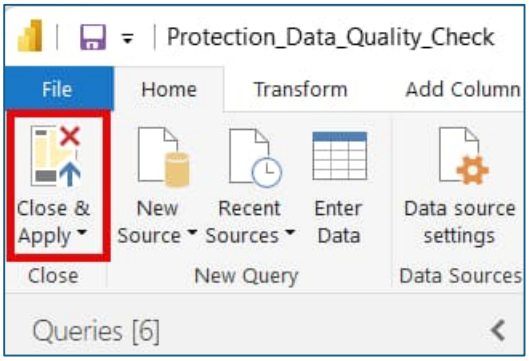


Figure 14: Close & Apply to read CSV files and populate visuals.

6.2 Line/XFMR terminal summary

Navigate to the **Line/XFMR Terminal Summary** page, using the navigation bar at the bottom of the report. See Figure 15.

Figure 15: Line/XFMR terminal summary page.

This page shows the data quality issues found at the level of a line or transformer (XFMR) terminal and can also be considered an asset-level summary. The types of issues identified, and the severity level of those issues are shown in Table 1.

Table 1: Asset-level data quality issues and severity.

No.	Type of data quality issue	Severity
1	LZOP missing for the line terminal or transformer	High
2	LZOP logic missing (if LZOP is present)	High
3	LZOP present but relays are missing	High
4	Protective elements missing contact logic code	High
5	Protective elements with CT or VT issues	High
6	Duplicate element-level contact logic codes	Low
7	Elements found with Trip CB flag = “Y” but no element-level contact logic code	Low
8	No data quality issues	None

Figure 16 summarizes the number of issues found at the asset level. Note that these counts are high-level counts, and do not report the number of instances of that issue for that asset. For example, multiple elements in the LZOP may show CT connection problems, but they are all counted as one CT issue, when reported at the asset level. Counts of instances of issues found at the element level are available in another report described in a later section.

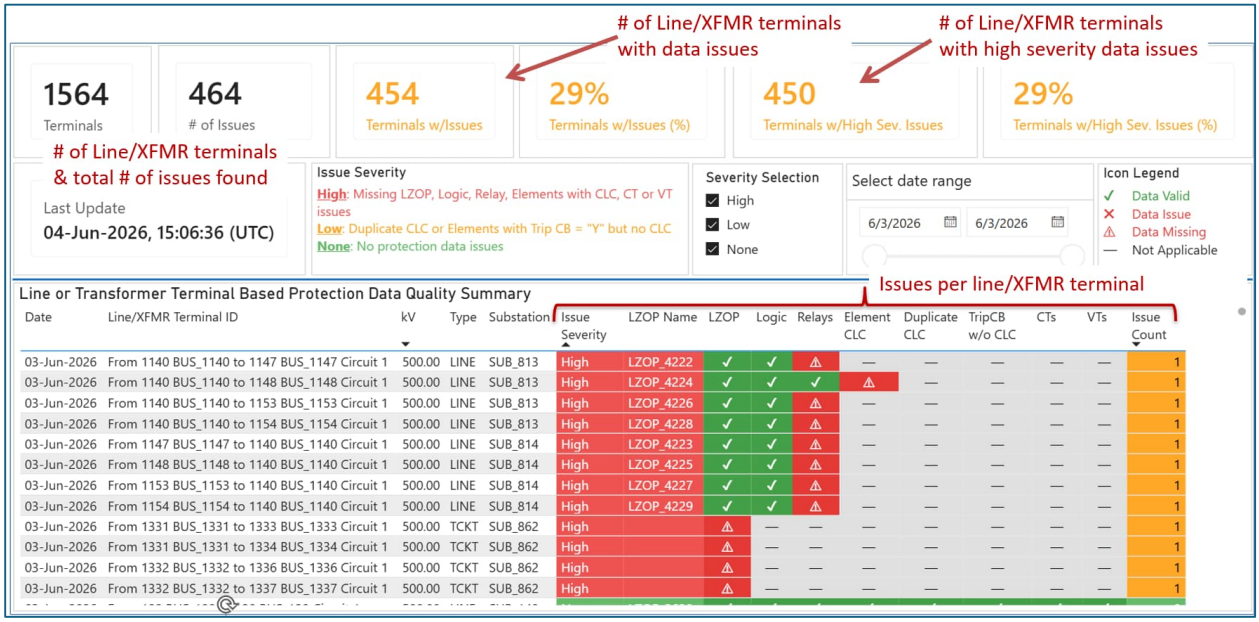


Figure 16: Line/XFMR terminal data quality issues.

In this example, data for 1564 line and transformer terminals are reported. The study found 464 asset-level issues, spread out among 454 different terminals, which constitutes approximately 29% of terminals with one or more quality issues. 450 out of the 454 terminals with data quality problems exhibit **high** severity issues.

The information in the visual can be filtered according to type of severity, or date when the data issues were captured by the APA macro.

Within the visual, different icons such as a green check mark, red x mark or red triangle with exclamation mark (see Icon Legend above the table visual) indicate the type of quality issue discovered for each asset.

Below the table visual, the report shows the distribution of the various issues found according to the voltage (kV) levels found in the data set. See Figure 17 and Figure 18.

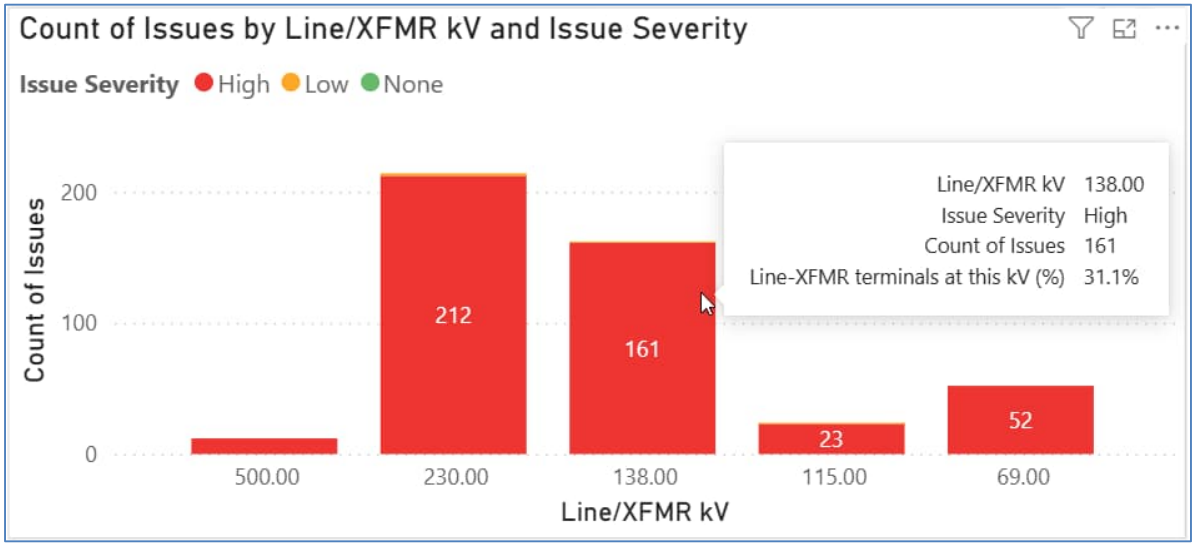


Figure 17: Data quality issue distribution by kV.

Per Figure 17, 212 high-severity issues were found at the 230 kV voltage level. There are some low-severity issues as well, but not visible at this zoom-level. The severity selection filter can be used to highlight them.

Figure 18 shows the number of assets at different voltage levels that exhibit high-severity, low-severity or no data quality issues. If an asset exhibits both high- and low-severity issues, it is counted as a high-severity case.

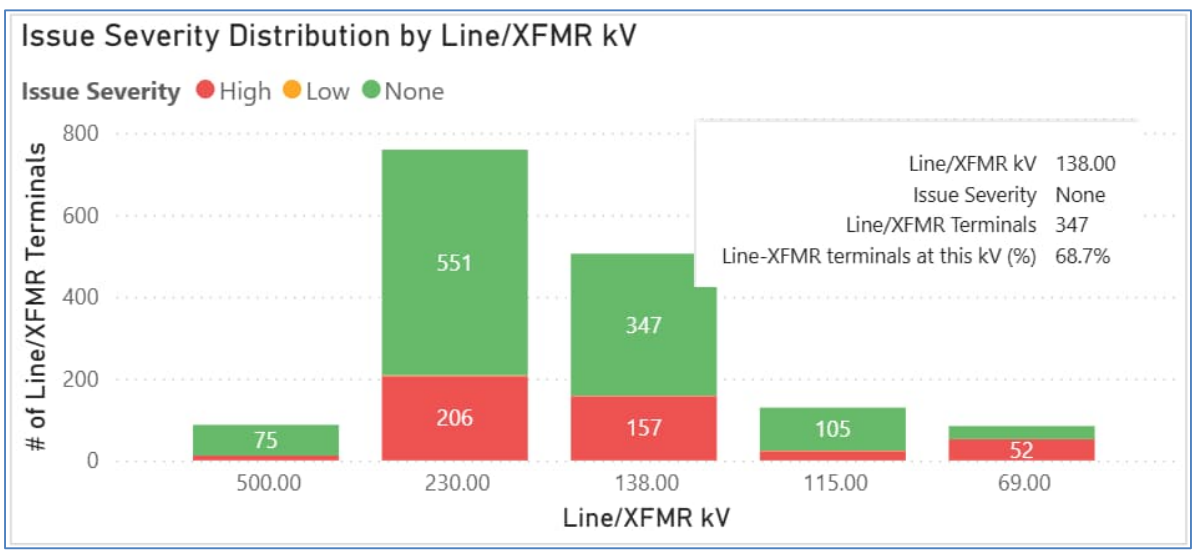


Figure 18: Distribution of severity of issues by line or transformer terminal.

Additional details for each high- and low-severity issue type are also shown on this report page. See Figure 19.

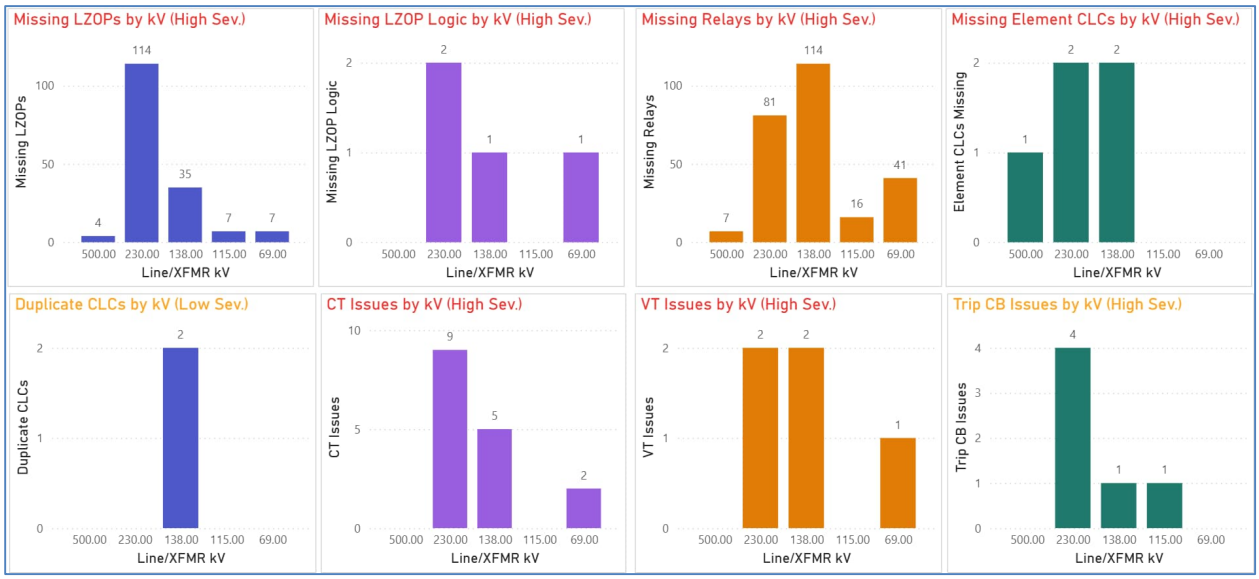


Figure 19: Distribution of high- and low-severity issue types by kV.

It is possible to open the LZOP data form in Gridscale X APA from the Power BI table visual. Scroll the data table shown in Figure 16 full to the right, as shown in Figure 20.

1. Right-click the cell where the text **VIEW_LZOP_DATA xxxx** appears and choose **Copy | Copy value**.
2. The number after VIEW_LZOP_DATA is the LZOP tag, a unique internal number that APA assigns to each LZOP the user has created in the database.

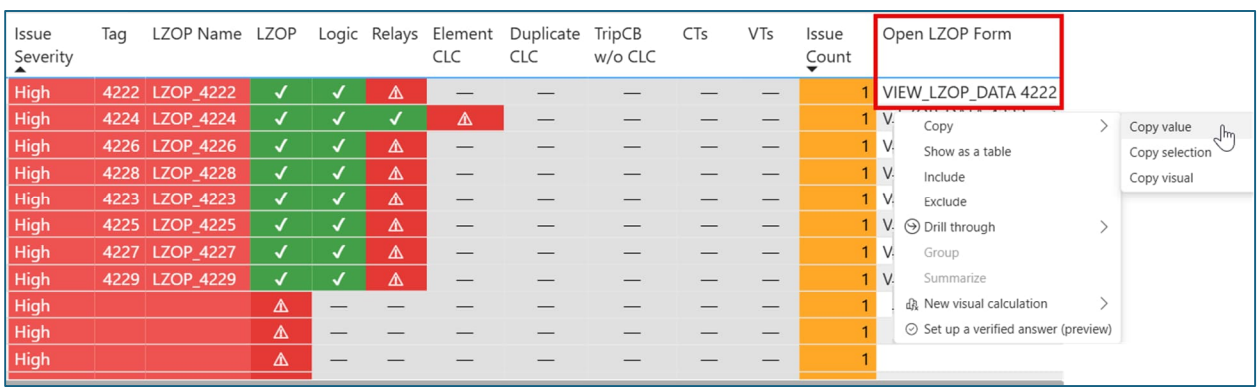


Figure 20: CUPL command to open the LZOP data form.

3. The copied command can be pasted directly into the command line in APA to open the LZOP data form.
4. You must have APA open, with appropriate database attached, etc. See Figure 21.

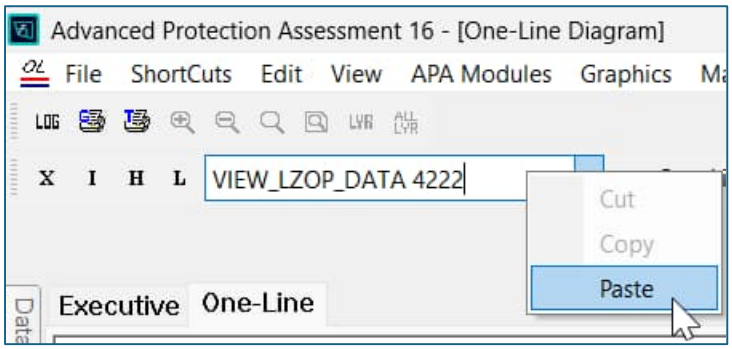


Figure 21: Paste CUPL command in the APA command line and press **Return**.

Finally, the line/xfmr terminal summary table allows you to drill through to the element-level data, as shown in Figure 22.

1. Right-click on any row (with relays and elements) and choose **Drill through | Element-Level Details**.

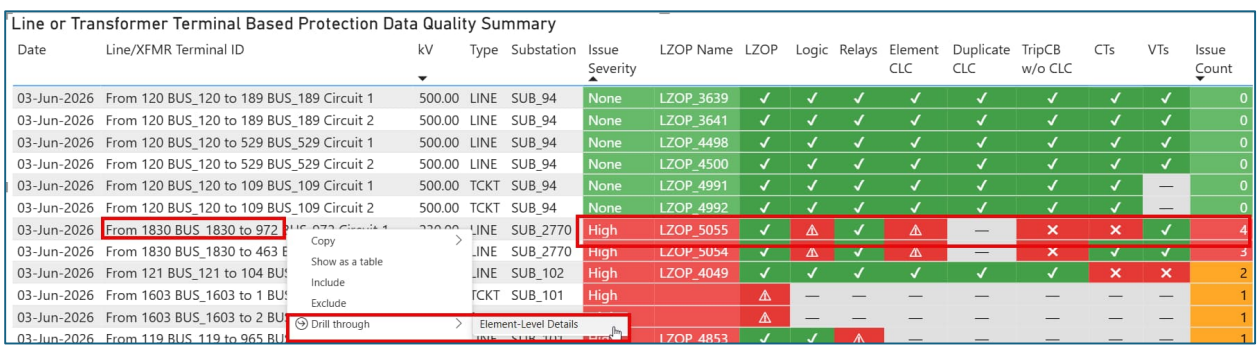
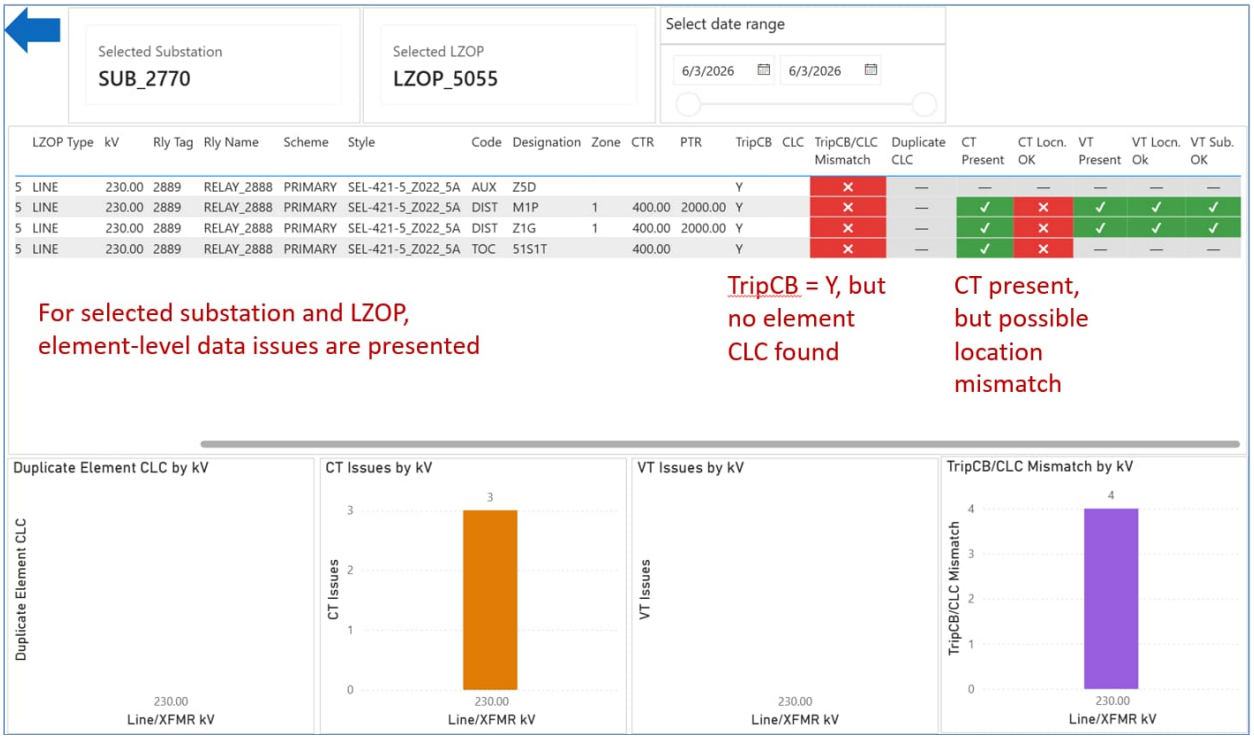


Figure 22: Drill-through to element-level details.

This action will take you to the **Element-Level Details** page of the report.

6.3 Element-Level details page

Figure 23 shows details for the selected asset (based on substation and LZOP). There are four elements in the relay where the Trip CB flag = "Y" or element contact logic code is not empty. Suitable icons (green check mark, red x mark) indicate valid or invalid data.

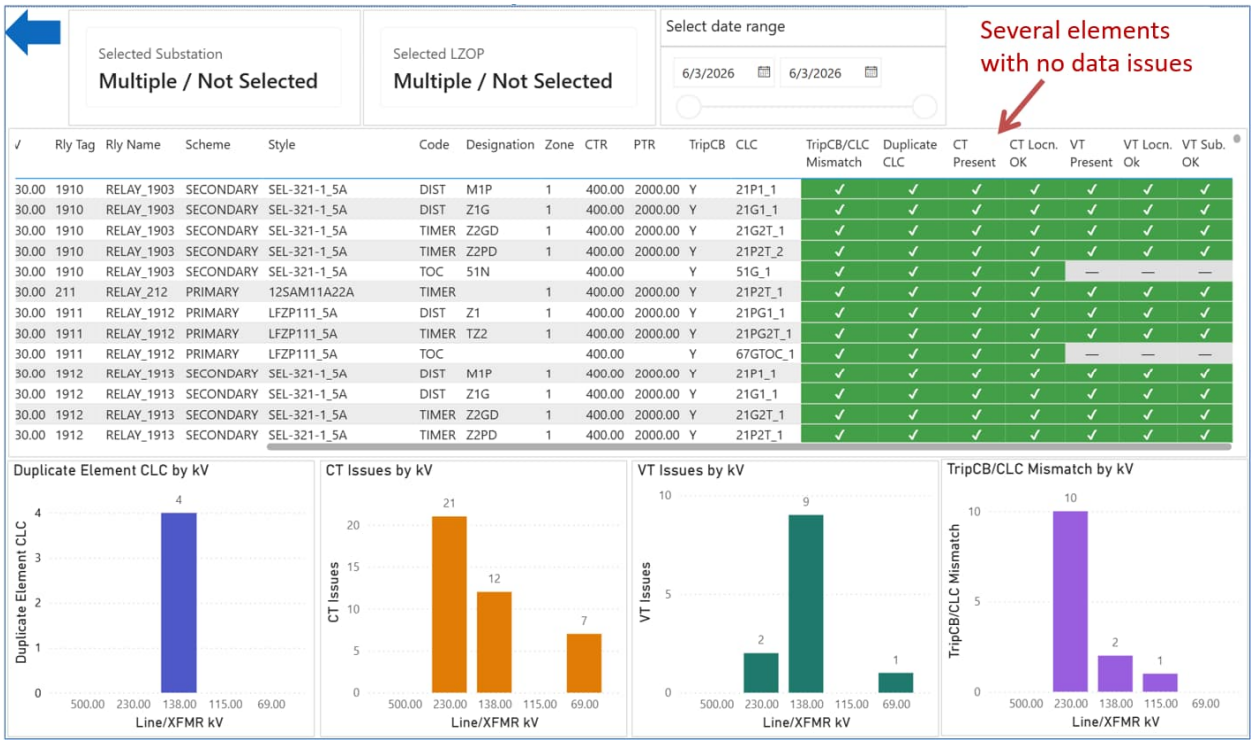


	LZOP Type	kV	Rly Tag	Rly Name	Scheme	Style	Code	Designation	Zone	CTR	PTR	TripCB	CLC	TripCB/CLC Mismatch	Duplicate CLC	CT Present	CT Locn. OK	VT Present	VT Locn. Ok	VT Sub. OK
5	LINE	230.00	2889	RELAY_2888	PRIMARY	SEL-421-5_Z022_5A	AUX	Z5D				Y		✗	—	—	—	—	—	—
5	LINE	230.00	2889	RELAY_2888	PRIMARY	SEL-421-5_Z022_5A	DIST	M1P	1	400.00	2000.00	Y		✗	—	✓	✗	✓	✓	✓
5	LINE	230.00	2889	RELAY_2888	PRIMARY	SEL-421-5_Z022_5A	DIST	Z1G	1	400.00	2000.00	Y		✗	—	✓	✗	✓	✓	✓
5	LINE	230.00	2889	RELAY_2888	PRIMARY	SEL-421-5_Z022_5A	TOC	51S1T		400.00		Y		✗	—	✓	✗	—	—	—

Figure 23: Element-level details for selected asset.

In this example, we find four elements where the trip CB flag = "Y", but there are no contact logic codes on the elements, which are necessary for tripping to take place. Also, three of the elements exhibit potential CT location issues, even though CTs are present and connected. The lower portion of the visual indicates the count of issues by voltage level.

Clearing the filters (substation and LZOP) in this visual will expand the table to show all the relay elements that were evaluated by the APA macro. See Figure 24.

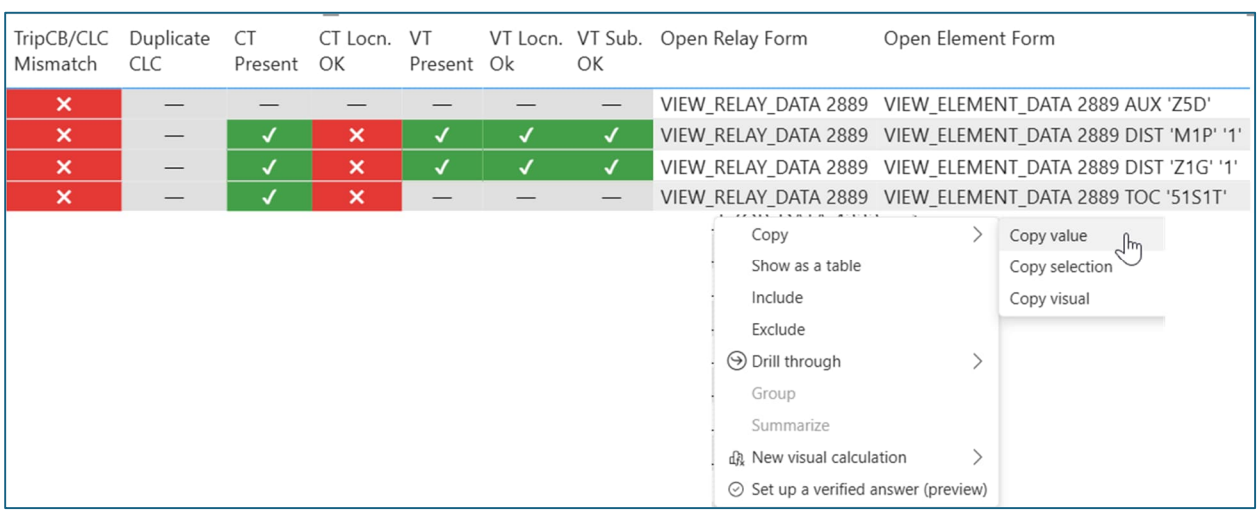


✓	Rly Tag	Rly Name	Scheme	Style	Code	Designation	Zone	CTR	PTR	TripCB	CLC	TripCB/CLC Mismatch	Duplicate CLC	CT Present	CT Locn. OK	VT Present	VT Locn. Ok	VT Sub. OK
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	DIST	M1P	1	400.00	2000.00	Y	21P1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	DIST	Z1G	1	400.00	2000.00	Y	21G1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	TIMER	Z2GD	1	400.00	2000.00	Y	21G2T_1	✓	✓	✓	✓	✓	✓	✓
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	TIMER	Z2PD	1	400.00	2000.00	Y	21P2T_2	✓	✓	✓	✓	✓	✓	✓
30.00	1910	RELAY_1903	SECONDARY	SEL-321-1_5A	TOC	51N		400.00		Y	51G_1	✓	✓	✓	✓	—	—	—
30.00	211	RELAY_212	PRIMARY	12SAM11A22A	TIMER		1	400.00	2000.00	Y	21P2T_1	✓	✓	✓	✓	✓	✓	✓
30.00	1911	RELAY_1912	PRIMARY	LFZP111_5A	DIST	Z1	1	400.00	2000.00	Y	21PG1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1911	RELAY_1912	PRIMARY	LFZP111_5A	TIMER	TZ2	1	400.00	2000.00	Y	21PG2T_1	✓	✓	✓	✓	✓	✓	✓
30.00	1911	RELAY_1912	PRIMARY	LFZP111_5A	TOC			400.00		Y	67GTOC_1	✓	✓	✓	✓	—	—	—
30.00	1912	RELAY_1913	SECONDARY	SEL-321-1_5A	DIST	M1P	1	400.00	2000.00	Y	21P1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1912	RELAY_1913	SECONDARY	SEL-321-1_5A	DIST	Z1G	1	400.00	2000.00	Y	21G1_1	✓	✓	✓	✓	✓	✓	✓
30.00	1912	RELAY_1913	SECONDARY	SEL-321-1_5A	TIMER	Z2GD	1	400.00	2000.00	Y	21G2T_1	✓	✓	✓	✓	✓	✓	✓
30.00	1912	RELAY_1913	SECONDARY	SEL-321-1_5A	TIMER	Z2PD	1	400.00	2000.00	Y	21P2T_1	✓	✓	✓	✓	✓	✓	✓

Figure 24: Unfiltered element-level details.

As in the case of the LZOPs, it is possible to open the protective device data form or the element data form for any relay or element shown in the element-level details table.

1. Scroll fully to the right to view the CUPL commands needed to open the relay or element forms. See Figure 25.



TripCB/CLC Mismatch	Duplicate CLC	CT Present	CT Locn. OK	VT Present	VT Locn. Ok	VT Sub. OK	Open Relay Form	Open Element Form
✗	—	—	—	—	—	—	VIEW_RELAY_DATA 2889	VIEW_ELEMENT_DATA 2889 AUX 'Z5D'
✗	—	✓	✗	✓	✓	✓	VIEW_RELAY_DATA 2889	VIEW_ELEMENT_DATA 2889 DIST 'M1P' '1'
✗	—	✓	✗	✓	✓	✓	VIEW_RELAY_DATA 2889	VIEW_ELEMENT_DATA 2889 DIST 'Z1G' '1'
✗	—	✓	✗	—	—	—	VIEW_RELAY_DATA 2889	VIEW_ELEMENT_DATA 2889 TOC '51S1T'

Figure 25: CUPL command to open protective device data or element data form in APA.

2. Copy the value of the cell and paste it into the command line in APA to open the appropriate data form.
3. The number shown after **VIEW_RELAY_DATA** is the relay tag, a unique internal number assigned by APA to each relay that you have created in the APA database.
4. The information shown after **VIEW_ELEMENT_DATA** includes the relay tag, element code (IOC, TOC, DIST, AUX, TIMER, etc.), the element designation and the zone or unit if applicable (DIST, TIMER).

7 Troubleshooting

If you encounter issues, try the checks below.

❌ **No Data Appears in Power BI**

- Verify that the **APA_User_Dir** parameter is correct and matches your installation of APA
- Ensure CSV files exist in:
  - **PPDCR_Output_Files\Sub_Output**
  - **PPDCR_Output_Files\Ele_Output**
- Confirm you clicked **Close & Apply** after updating the **APA_User_Dir** parameter

❌ **CSV Files Not Found**

- Confirm the macro ran successfully in APA
- Verify the macro files are:
  - correctly copied
  - correctly referenced in **cape.cfg**

❌ **Macro Not Visible in ShortCuts Menu**

- Verify:
  - **cape.cfg** was updated correctly
  - **ol_shortcuts.txt** contains the correct entries
- Restart APA after making changes

❌ **APA Command Column is Blank**

- The selected asset may not have a valid LZOP assigned
- This is expected for certain data conditions (missing LZOP for an asset)

❌ **Errors During Power BI Refresh**

- Check that the folder path exists and is accessible
- Ensure CSV files are not open in another application (e.g., Excel)
- Confirm file naming and folder structure have not been altered

8 Planned improvements

Several improvements are in the roadmap to identify additional data quality issues via the APA macro. As explained before, CUPL does not have access to several internal data structures inside APA that would make it easier to identify data problems. As such, the macro will have to trawl through multiple data tables in the database to capture more complicated issues. The planned improvements are:

1. Detection of CT issues for current differential (CDIFF) elements
2. Detection of polarizing CT and VT issues for directional (DIR) elements
3. Detection of CT summation point issues such as missing CT parents
4. Detection of mismatches between element contact logic codes and tokens used in the Boolean LZOP logic expressions
5. Detection of issues in teleprotection (pilot) schemes

As updates are made and new versions are available, Grid Integrity Solutions will inform you via email. Please respond to the announcement to receive the updated files.

You can also obtain the latest version from the GIT repository at https://github.com/Grid-Integrity-Solutions/protection-qc.

9 Technical support

Please contact Grid Integrity Solutions at info@gridintegritysolutions.com to report bugs, request technical support or provide suggestions for improvements and updates.